1. Introduction to Ergonomics

What is Ergonomics?

In early times, man needed to use only simple tools to scratch out a living from the earth. With the coming of the industrial revolution, the complexity and number of tools that man must use in the workplace has increased astronomically. No longer is man subjected to only the natural environmental conditions. Workers must work in an industrial atmosphere where they are constantly subjected to conditions that may cause damage to their health and physical well being. Machinery in the workplace can cause noise and vibration as well as contamination of the atmosphere with toxic materials. In addition, the typical industrial environment is often not a serene atmosphere in which to work. The worker is bombarded by many auditory and visual stimuli and must react to these stimuli to perform tasks. Within this environment, the tasks that must be performed include both physical and mental activities. It is obvious that the stress placed on the worker is significant and that proper design is required to minimize the strain placed on the worker.

Ergonomics, though a recently identified discipline of study, has its basis in the study of Ramazzini who, in 1700, discussed the ill-effects of poor posture and poorly designed tools on the health of the worker. The term, "ergonomics," is derived from the Greek words "ergos," meaning work, and "nomikos," meaning law. Ergonomics is, then, the study of work laws.

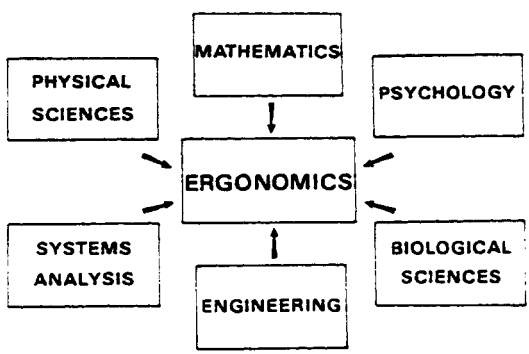
Some confusion exists concerning the field of ergonomics and the objectives toward which it is directed. Part of the confusion is a result of the fact that two general terms are used for essentially the same discipline. The term "ergonomics," which is used more widely in England, and the term "human factors," which is prevalent in use in the United States, essentially describe the same discipline or field of study. It might be argued that the field of human factors in the United States is broader in its applicability since it involves the design of products for the consumer market place as well as the design of work-related tasks and equipment. However, generally the major application of both fields is toward the use of bioengineering and biomechanics to improve the workplace environment for the worker.

Ergonomics draws from many fields or disciplines of study. Figure 7.1.1 illustrates the major disciplines from which ergonomics draws. In the physical sciences, both physics and chemistry provide basic information that is used in the field of ergonomics. Mathematics, through the use of statistics and biometrics (the measurement of body structure), is also widely applied in the study of work laws. The biological sciences give to the field

of ergonomics the studies of anatomy, physiology, and anthropometry. Psychology provides a basis for learning theory as well as human reaction to various stimuli. The field of engineering is important in the proper design of equipment for use by humans in the workplace. Finally, the field of systems analysis provides a basic structure whereby the analysis of the job can be performed.

Figure 7.1.1

Major disciplines.



The basic objective of ergonomics is to obtain an optimum relationship between the worker and the work environment. The worker, in performing a job, must act to complete assigned tasks in a manner that will result in the desired product being generated in the most efficient manner possible. At the same time, the worker must be provided with protection against undue physical, biological, and psychological strain that might occur as a result of the performance of the required tasks.

In performing the tasks assigned, the worker is influenced by a number of factors that must be considered in attempting to optimize the relationship between the worker and the job. Among the important factors that must be considered are the thermal conditions in the workplace, the amount of illumination present, the physical and mental requirements of the job, the noise level in the workplace, the worker's exposure to hazardous materials, and the interaction between the worker and the equipment that is required for the job.

Consider the worker who is working on an assembly line. This worker must perform the assigned tasks in a manner so as to produce the desired quality within a specified time period. If the worker does not perform the tasks properly, then the product will be rejected, resulting in a loss to the company of not only the materials in the product but also the cost of any other workers' time invested in the product to the point of rejection. If the worker does not perform the required tasks in the allowable time frame, the

assembly line will eventually slow to the pace of the worker, lowering the output and raising unit cost.

However, there are many factors that may act to affect the performance of the worker. For example, the assembly line may be located in an area of relatively high heat load, thus exposing the worker to thermal stress. At a minimum, this thermal stress can place a physiological strain on the worker, resulting in potential illness. The absence of adequate illumination can result in improper performance of the assigned tasks and a potential strain to the worker's vision. Inadequate illumination can also be a factor in accidents, resulting in potential injury to the worker.

Automation has had a major influence on the work that is performed in industry today. It has resulted in the elimination of many of the heavy physical tasks that must be performed. However, in spite of automation there are still many jobs that require physical tasks to be performed. The physical demands, though in many cases different in nature from those previously experienced in the workplace, can result in a stress being placed on the individual worker while performing a job. This stress can affect the worker's performance as well as his health and safety.

Perhaps one of the largest changes that has occurred in the industrial environment today when compared to the industrial environment of the past isthe increased mental demands placed on the worker. Workers must be constantly monitoring and controlling equipment while making decisions related to the actions that are necessary. These mental demands can cause fatigue in the worker just as the more physical tasks of the past caused fatigue.

Another factor that is more important in today's industrial workplace than it was in the past is the necessity for workers to interact with mechanical and electrical equipment. During this interaction, the potential for accident and injury can exist. The worker must maintain vigilance and use appropriate procedures to assure his safety and health. Also, as a result of the use of mechanical and electrical equipment, noise and vibration are generated in the workplace. This noise and vibration can have a physiological as well as an injurious effect on the worker. Noise itself can make communication difficult while, at a high exposure level, both noise and vibration may be physically harmful to the human body.

With the many chemicals that are currently in use in industry today, workers are potentially exposed to hazardous materials during work. These hazardous materials can cause both acute and long-time injury or illness to the exposed worker.

As is evident from the above discussion, the worker in the industrial environment of today is subjected to many potentially harmful factors. It is the job of the industrial hygiene engineer to assure that these factors do not result in injury and illness to the worker. The field of ergonomics or human factors is involved with designing the workplace and job in such a manner that the potential harmful effects of these factors in terms of health, safety, and efficiency are minimized.

The Man/Machine System

Let's look at the workplace to determine those major characteristics that are present. First, the workplace will include various units of mechanical and electrical equipment. The typical industrial workplace involves the use of such equipment as power presses, lathes, heat-treating equipment, and power saws. The nonindustrial work environment is not without the presence of mechanical and electrical equipment. Included in the nonindustrial workplace (e.g., office) is such equipment as typewriters, copy machines, and electronic data processing equipment. A second characteristic of most workplaces is the fact that manual tools are often required to perform the job. The manual tools can include hand power equipment, hammers, screwdrivers, and material handling equipment.

The third characteristic of the workplace is related to the jobs that are performed. Often these jobs are described by standard sets of procedures that are to be used to produce a given output. These procedures may require strict adherence, or they may be subject to worker interpretation, depending upon the type of job that is being performed.

The fourth characteristic of the workplace is that it is usually necessary for humans to act as operators or performers of tasks within the workplace environment. The workers interact with the equipment to perform many of the tasks that are involved in producing the output.

Finally, the workplace must be serviced by various facilities. These facilities include such things as light, heat, water, and waste removal that are required for the work processes being carried out in the workplace.

Now let's look at the worker's functions within the workplace as the interface with the various other characteristics or components. First, the worker acts to perform various physical tasks that are necessary to complete the job. These tasks might include such things as lifting and carrying materials and finished products, positioning materials for work operations, and using manual and power tools to perform the work functions.

A second function of the worker within the workplace is to act to control the processes that are performed by machines. The worker initiates the action that starts the machines to work and controls the processes so that the job is carried to completion properly. Finally, the worker acts to stop the process should something in the system fail, or at the completion of the process. In order to control the equipment, it is necessary that the worker monitor the process being performed. This monitoring may be done on the work being performed by others, such as is the case with an inspector; or the monitoring may be performed by the machine operator.

Finally, in addition to performing physical tasks and monitoring the work processes, the worker must also share the environment with the machines in the workplace. As a result, the worker is subjected to the potential hazards that exist as a result of the use of mechanical and electrical equipment to perform the job.

Ergonomics is concerned with this interface between the worker and the job. It is concerned with the physical and mental demands on the worker that are necessary to perform the job. Also, ergonomics is concerned with the manual tools that are used to perform the job and the mechanical equipment that must be monitored and controlled by the worker.

Finally ergonomics is concerned with the work space in which the worker must perform his tasks.

More specifically, then, ergonomics is concerned with the following:

1. The design of tools that are used in order to match the physical characteristics of the worker with the functioning of the tools.

2. The design of the workplace space itself to meet the physical characteristics of the worker.

3. The analysis and design of controls and displays to allow the worker to operate and monitor processes efficiently with minimum error.

4. The development of job procedures that meet the capabilities of the worker in the system.

5. The minimization of external forces that can act to affect the worker in the workplace. These external forces include such things as noise, thermal conditions, illumination, and vibration.

The Systems Approach

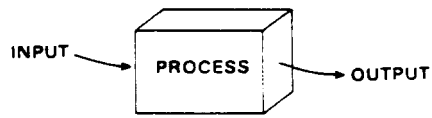
Since we are concerned with the interface between the worker and the work environment, it is necessary that a logical procedure be developed to analyze this interface. The systems approach provides a method for investigating the worker/machine system in the work environment in such a logical manner. Using the systems approach, the total system is analyzed by dividing it into manageable subparts that can then be analyzed for the factors of concern.

The systems approach commonly uses a method that is called the "black box" method. In this approach, the major components of the system are considered to be the input, processes, and output of the system. In the workplace system, the input is in terms of the raw materials, data and information that are necessary to perform the job. The process involves the steps that are taken to convert the input to the desired output. The process may be many things such as the heat treating of metals, the multi-step fabrication of metal products, the manufacture of chemicals, or the processing of information that occurs in a data processing facility. Whatever the process, it results in a desired output, or product.

Using this black box approach, the first step is to define the desired output or purpose of the system. What output do we desire? What characteristics should the output have? What tolerances or error will be permitted? Questions such as these help to define what should be accomplished during the work process.

Figure 7.1.2

The black box.



The second step of this approach is to define the inputs that are required to obtain the desired outputs. What materials are required to produce our product? What skills and knowledge are required of the worker to accomplish the desired result? What constraints are present within the system? These constraints may be environmental or mechanical factors, capital cost, or the need for adequately trained personnel.

Finally, after the outputs and inputs have been defined, the process itself can be described. How can the input be converted into the desired output, given the constraints that are present? What functions are required to convert the inputs into the outputs? What tasks are required to do the desired job?

Function and Task Analysis. One method for defining the process is through the use of function and task analysis. A job or group of jobs can be divided into functions and tasks or activities. These functions and tasks can then be analyzed. Job functions are themselves broad groups of activities that help to accomplish the objective or purpose of the organization. As an example of a function, consider driving an automobile. One function that is required is the observation of the road and surrounding conditions to determine what changes in operation are necessary. A second function is the steering of the automobile in order to assure the appropriate direction is being maintained. A third function includes acceleration and braking of the automobile to maintain the desired progress.

In some cases, higher order groupings of phases or stages is helpful. Such functional analysis is useful in the initial design of new processes and procedures. This approach has been commonly used in the space industry to define the various stages, phases, or functions within a mission.

A common set of functions that are useful in the initial analysis of a job is:

1. Preparation--What steps are necessary to prepare the worker to perform the desired job?

2. Observation--What information and data must the worker have in order to perform the job?

3. Control--What steps must the worker take to control the processes involved? These steps include the mental processes or decisions that must be made to activate given controls.

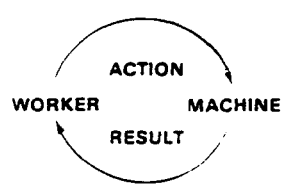
4. Physical Demands--What physical tasks must the worker perform to accomplish the desired objective?

5. Termination--What steps must be taken to terminate the job? This includes the cleanup after the job has been completed.

Functions 2 and 3 above are commonly considered in the feedback loop. In the feedback loop, the worker is presented with certain information concerning the status of the process. Using this information, the worker decides what control must be exerted on the process to obtain the desired result and institutes this control. Once the control is instituted, the process changes and the information concerning th process change are fed back to the worker to enable evaluation of the outcome of the changes. This feedback loop exists potentially in every job. It is not quite so obvious in some jobs as it isin others; however, in all jobs, such a feedback mechanism must exist in order for the worker to progress satisfactorily.

Figure 7.1.3

Feedback loop.



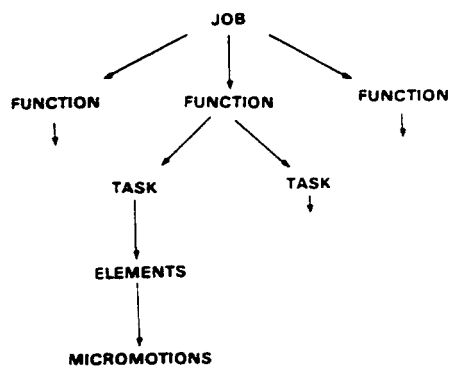
After the job has been divided into its major functions, each function can then be subdivided into tasks or activities. These tasks define the smaller steps that must be completed in order to obtain the desired output of the job. For example, consider the operation of a small business. The functions that may be performed within such a business include marketing, production, and accounting. Within the accounting function itself, further functionalization exists. These functions may include such things as accounts payable, accounts receivable, and payroll. Within the accounts receivable function, various tasks can be identified such as mailing invoices and maintaining a current status of unpaid invoices.

An even further breakdown of these tasks can be accomplished by breaking down each task into elements. These elements are the smallest activities that can be performed without considering simple psychomotor motions. For example,

an element of the task of mailing invoices might be the stuffing of envelopes with the invoices. Finally, the elements themselves can be analyzed in terms of the micromotions that are necessary to perform the particular element involved. Such analysis of micromotions is typical of the work done by Gilbreth and Maynard in the study of industrial engineering.

Figure 7.1.4

Job analysis.



The purpose of the function and task analysis is to divide the job(s) into small, manageable segments that can be analyzed. How far the job must be divided is dependent upon the desired outcome. If a job is being analyzed to increase efficiency or to remove potential hazards, it may be necessary to go to the level of micromotions. On the other hand, if the objective is to outline the specific duties that must be performed, the task level may be sufficient.

In any case the tasks, elements, or micromotions can be analyzed. This analysis involves determining such things as:

1. The man/machine interfaces.
2. The initiating conditions.
3. The required actions by the worker.
4. The decision rules and required feedback.
5. The potential errors.
6. The time required to perform the task or element of interest.
7. The potential harmful exposures.
8. The work station requirements.
9. The required skills and knowledge of the worker.

This approach can be used for the identification of many types of hazards that might exist within a given job. It can also be used to redesign existing jobs or to create designs for new jobs. Thus, it is both an identification and control technique that can be used by the industrial hygiene engineer.

Procedure for Conducting a Functional and Task Analysis. The first step that must be performed is to subdivide the job into the major functions that are necessary to accomplish the desired output. Remember the five general functions or stages as the beginning point (i.e., the preparation stage, the observation stage, the control stage, the physical requirement stage, and the termination stage).

The second step is to divide the functions into the various tasks that must be performed. Again, the five stages may be helpful in breaking each of the major functions down into tasks.

For each task itself, it may be necessary to break it down further into the elements and/or micromotions that are necessary, depending upon the desired analysis. The five stages--preparation, observation, control physical requirements, and termination--may help to identify the groupings of elements within the task itself, since it is likely that most of these stages will be present in a given task. Once the desired lowest level has been attained, the analysis can begin. This analysis involves asking a number of questions about each of the individual tasks, elements, or micromotions that are being analyzed. These questions include:

1. What are the initiating conditions? What is the cause or stimulus that results in the task being performed in the first place?

2. What actions must be carried out? What steps must be performed by the individual and/or machine to accomplish the desired result?

3. What feedback is required to assure that the results of the action are as desired?

4. What potential errors are possible? What is the cost of these errors occurring in terms of damage to equipment, materials, and/or the operator's physical well being?

5. What hazards are present that can cause illness or injury to the worker?

6. What is the required reaction time necessary to initiate the task?

7. What is the time frame in which the task must be completed?

8. What tools and equipment are required to complete the task or element?

9. Where is the task or element being performed? What is the physical location and structure of the workplace in which the task or element is performed?

10. What physical demands are placed upon the worker to perform the task or element?

11. What skills and knowledge are required of the worker to perform the task or element?

The objective of the analysis of the task or element is to define the task or element in such a way that an analysis can be made to determine the changes that can be introduced to that task or element to meet the desired result. This is done with the objective of also lowering the potential stress on the individual performing the task. If an existing job is being analyzed and the above questions have determined the current requirements of the job, it is desirable to answer some key questions that might help to obtain our objective.

First, we might want to consider if the physical demands of the task or element can be lowered. Is it necessary that the worker perform the job as it is currently outlined? Can equipment replace some of the physical demands on the worker? Can the worker be seated rather than standing to perform the task? Does the worker have to perform this task or element in close proximity to the mechanical process?

Second, can equipment be substituted to replace the operator totally? Can the entire process be automated to remove the worker from the workplace? Such action will eliminate the need for further consideration of potential strain on the worker.

Another question that might be asked is, "Does the equipment being used require redesign?" Are there potential hazards that are characteristic of the equipment? If so, can these potential hazards be designed out of the system by modifying or substituting for the equipment.

Can the workplace itself be changed? Can modifications be made to the work space, controls, displays, tools and equipment being used to allow for more efficient operation?

Finally, can the task or element procedures be changed to lower the potential for critical errors occurring? The removal of these critical errors will allow for a more efficient and less costly operation, while at the same time providing a safer environment for the worker.

Using functional and task analysis, certain benefits can be obtained. By analyzing the tasks and elements, the job, workplace, and tools can be designed to fit the worker's characteristics. The result is that there will be less possibility of injury or accident as well as lowering the fatigue occurring in the worker. Additional benefits are obtained by using this approach. These include improved production, lower costs, improved morale, and improved manpower utilization.

Summary

Ergonomics is the study of the worker's function within the workplace. The objective of ergonomics is to fit the worker and the job together in such a manner as to create a total system that is efficient and safe in producing the desired output. In order to begin an ergonomic study of a particular job, it is desirable that a functional and task analysis be conducted. This functional and task analysis provides a basis by which the job can be broken down into small segments that can then be analyzed for the critical components. After analyzing these components, the job can be designed in such a manner as to produce an efficient and safe worker/machine interface.